AgriDirect AI

Crop demand prediction · Direct farmer-to-company marketplace

PRODUCT REQUIREMENTS DOCUMENT

Product Requirements Document

AI crop demand prediction + direct farmer-to-company marketplace

Version	**1.1 (revised from v1.0 draft, 13 Oct 2026)**
Date	**24 October 2026**
Original author	**Onkar (Project Lead / Client Liaison)**
Revised by	**Development team**
Status	**Baseline for development — open items in Section 14**

Confidential — prepared for the AgriDirect AI development team.

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1. Revision summary

Version 1.1 keeps every requirement from the v1.0 draft and adds what a developer needs before writing code: user personas, user stories with acceptance criteria, a screen inventory, field-level validation rules, a phased demo plan, and clarified priorities. Nothing in v1.0 has been removed.

Area	Change in v1.1
Personas	Added three personas (farmer, company buyer, admin) to anchor UX decisions.
User stories	Each Must-have requirement now has one or more stories with testable acceptance criteria.
Screen inventory	New Section 8 lists every screen, its route and which FR it satisfies.
Validation rules	New Section 9 defines field formats, units and limits (quantity, phone, GST, dates).
Phase 0 prototype	Added a clickable frontend prototype with mock data as a deliverable before backend work (Section 12).
Deal statuses	State machine made explicit, with allowed transitions (Section 7.4).
Partial fulfilment	Clarified: a listing can be split across deals; remaining quantity stays live (from worked example).
Languages	English ships first in the prototype; Hindi and Marathi remain MVP Must-haves.

2. Product overview

AgriDirect AI helps farmers decide **what to grow** and then **who to sell to**. It combines two products in one app:

- **AI Crop Demand Predictor** — uses mandi price/arrival data, weather and soil to estimate local demand level, demand trend and price direction for a crop, and ranks suitable crops for a farmer's location and season.
- **Direct Farmer-to-Company Marketplace** — farmers list expected or ready produce; verified companies post requirements; a matching engine connects them; both sides negotiate and confirm a deal on the platform.

> **Positioning line:** We don't only predict what farmers should grow — we connect them with who needs it.

2.1 Core flow

#	Stage	What happens
1	Farmer input	Farmer registers and enters location, land area, crop, expected quantity and harvest date.
2	Data signals	System pulls weather, soil and mandi price/arrival data for that location and season.
3	AI prediction	Model estimates demand level, demand trend and price direction with a confidence score.
4	Recommendation	Ranked shortlist of suitable crops with a plain-language explanation.
5	Matching	Produce listings matched to company requirements by crop, quantity, distance, date and quality.
6	Direct deal	Company sends enquiry/offer; parties negotiate and confirm the deal on the platform.

3. Problem statement

- Farmers choose crops with little visibility into local demand, mandi prices, weather and soil; information is scattered across mandis, traders and informal networks.
- When many farmers follow the same signal they plant the same crop, oversupply follows and harvest-time prices fall.
- After harvest, farmers rely on the trader / commission agent / mandi chain and have no easy way to find bulk buyers who actually need their produce.
- Food processors, exporters, retailers and feed makers struggle to find reliable nearby supply in the quantity and timeframe they need.
- Farmers need simple, timely guidance in their own language, not another complex dashboard.

4. Goals and success metrics

4.1 Product goals

- Give farmers a clear, easy-to-read crop and demand recommendation before sowing.
- Create a direct discovery channel between farmers and bulk buyers.
- Deliver an MVP demonstrable end-to-end in one pilot region.

4.2 MVP targets (to be confirmed with client)

Metric	Target	How measured
Price forecast quality	Beats naive baseline (last-value / seasonal-naive) on MAPE and RMSE	Time-based hold-out per region and season; reported in model card
Demand class quality	Macro-F1 above baseline; accuracy reported	Same hold-out; confusion matrix per crop
End-to-end flow	Listing → match → offer → accepted with no manual step	Staging demo script (Section 13)
Languages	English, Hindi, Marathi	All UI strings externalised; reviewer sign-off per language
Page performance	Key pages < 3 s on 4G; prediction API < 2 s (p95)	Lighthouse (Moto G-class throttling); API load test
Pilot	1 district in Maharashtra, 3–5 crops	Admin crop catalogue + live data
Early adoption (suggested)	50 farmers, 10 verified companies, 10 deals in pilot	Admin analytics

5. Users and personas

Role	Description	Key needs
Farmer	Smallholder or mid-size grower; may have low digital literacy; basic Android phone.	Simple UI, local language, crop advice, find buyers, safe deals.
Company / Buyer	Processor, trader, exporter, retailer, FPO or institution needing bulk produce.	Post requirements, discover nearby supply, compare listings, contact farmers.
Admin	Platform operator (client team).	Verify users, manage crops/data, monitor deals, analytics, disputes.
Data / ML engineer	Maintains pipelines and models.	Retrain, monitor accuracy, manage data sources.

5.1 Personas

Ramesh Patil — farmer, 42, Ahilyanagar district

Grows 4 acres of onion and maize with well irrigation. Uses WhatsApp and YouTube daily on a ₹9,000 Android phone. Reads Marathi comfortably, English poorly. Sold onion at a loss last year because "everyone planted onion". **Wants:** one clear suggestion before sowing and a buyer who pays on time.

Sneha Kulkarni — procurement manager, food processor, Pune

Buys 40–60 tonnes of maize and soybean a month for a feed unit. Today calls 6–8 traders and has no visibility on quality until delivery. **Wants:** to post a requirement once, see verified nearby supply with ready dates, and shortlist in minutes.

Platform admin — operations executive

Verifies company GST numbers, watches for fake listings, checks that the daily data pipeline ran. **Wants:** a queue of pending verifications, a deal monitor and a red/green data-pipeline panel.

6. Scope

6.1 In scope — MVP (Phase 1)

- Registration and login (farmer, company, admin) with OTP phone verification.
- Farmer profile, land and crop details; company profile with verification.
- Data ingestion: mandi prices/arrivals, weather, soil.
- AI demand and price-direction prediction plus ranked crop recommendation with explanation.
- Farmer produce listings and company requirement postings.
- Rule-based matching engine.
- In-platform enquiry → offer → negotiation → accept/reject deal flow, with notifications.
- Multilingual interface (English, Hindi, Marathi).
- Admin dashboard: verification, moderation, monitoring, basic analytics.

6.2 Out of scope for MVP (Phase 2+)

- Online payments, escrow, invoicing.
- Logistics booking and tracking.
- IoT, drone and satellite data.
- Government scheme / procurement integration; export market signals.
- Native iOS/Android apps (MVP is a responsive PWA).

- Contract farming, credit and insurance.

6.3 College demo scope (Phase 0 prototype)

> Before backend work, a **frontend-only clickable prototype** with realistic mock data covers every MVP screen (Section 8). It is used for the college presentation, client walkthroughs and usability testing with 3–5 farmers. It makes no real API calls, sends no OTP and stores nothing beyond the browser session.

7. Functional requirements

Priority: **M** = Must (MVP), **S** = Should, **C** = Could (later). IDs are unchanged from v1.0.

ID	Requirement	Pri.
FR-01	Register with mobile number + OTP; choose role (farmer / company). Email optional for companies.	M
FR-02	Farmer profile: name, village/district/state, GPS or map-pin location, land area, irrigation source, soil type (if known), preferred language.	M
FR-03	Company profile: name, type, GST/registration number, operating locations, contact person.	M
FR-04	Admin verifies or rejects companies (optionally farmers); only verified companies can post requirements.	M
FR-05	Farmer requests crop advice by location, land area and season; system returns ranked crops with demand level, trend and price direction.	M
FR-06	Every recommendation shows a plain-language explanation and a confidence indicator.	M
FR-07	Farmer creates a produce listing: crop, variety, quantity, unit, ready date, expected price (optional), location, photos (optional).	M
FR-08	Each listing shows AI insight: local demand, demand trend, aggregate nearby company requirement.	M
FR-09	Company posts a requirement: crop, quantity, required-by date, delivery location, quality spec, price offer or range.	M
FR-10	Matching engine ranks listings vs requirements by crop, quantity, distance, date and quality; both sides see suggestions.	M
FR-11	Company searches and filters produce (crop, quantity, district, ready date, distance) and contacts farmer or sends offer.	M
FR-12	Deal workflow with statuses and full history visible to both parties (Section 7.4).	M
FR-13	SMS / WhatsApp and in-app notifications for matches, offers and status changes.	S
FR-14	Multilingual UI and advisory text (EN/HI/MR), switchable any time.	M
FR-15	Market dashboard: price and arrival trend charts per crop and mandi.	S
FR-16	Voice input / read-aloud advisory.	C
FR-17	Ratings and reviews after deal completion.	S
FR-18	Admin dashboard: users, moderation, deal monitoring, data-pipeline status, model performance.	M
FR-19	Report/flag a listing or user; admin review workflow.	S
FR-20	Export deal history and listings to CSV/PDF.	C

7.1 User stories — farmer

Story	Acceptance criteria
As a farmer I want to log in with my phone and an OTP so I don't need a password.	6-digit OTP; expires in 5 min; max 3 sends per 15 min per number; wrong OTP 5 times locks for 30 min.
As a farmer I want to see which crops suit my land this season.	Shows 3–5 crops ranked; each card has demand level, trend arrow, price direction, confidence %, and one-sentence reason. Advisory disclaimer always visible.
As a farmer I want to list my maize so buyers can find me.	Listing saved with status Active; appears in company search within 1 minute; quantity in quintal or tonne; ready date ≥ today.
As a farmer I want to see which companies need my crop nearby.	Listing detail shows aggregate nearby requirement (e.g., 8 t within 50 km) and top 3 matches with score.
As a farmer I want to accept, counter or reject an offer.	Each action changes status per state machine; counter requires new price; history shows who did what and when.

7.2 User stories — company

Story	Acceptance criteria
As a buyer I want to register my company and get verified.	GST format validated; status Pending until admin action; cannot post requirements while Pending.
As a buyer I want to post a requirement.	Required: crop, quantity, unit, required-by, delivery location; price range min ≤ max.
As a buyer I want to search available produce near my unit.	Filters: crop, district, min quantity, ready before date, distance radius; results sorted by match score by default.
As a buyer I want to send an offer for part of a listing.	Offer quantity ≤ listing remaining quantity; on acceptance remaining quantity is reduced and listing stays Active if > 0.

7.3 User stories — admin

Story	Acceptance criteria
As an admin I want a queue of companies awaiting verification.	List shows name, GST, submitted date, documents; Approve/Reject with mandatory reason on reject; action logged in AuditLog.
As an admin I want to see if today's data pipeline ran.	Panel shows last run time and status per source (mandi, weather, soil); failed runs highlighted red.
As an admin I want to moderate flagged listings.	Flag queue with reason; actions: dismiss, hide listing, block user.

7.4 Deal state machine

From	Allowed next status	Who can trigger
— (new)	Enquiry, Offer sent	Company
Enquiry	Offer sent, Cancelled	Company; either side can cancel
Offer sent	Negotiation (counter), Accepted, Rejected, Cancelled	Farmer accepts/rejects/counters; company cancels
Negotiation	Negotiation (counter), Accepted, Rejected, Cancelled	Party receiving the latest counter
Accepted	Completed, Cancelled	Either party marks completed; cancellation needs reason
Rejected / Cancelled / Completed	— (terminal)	—

8. Screen inventory

These screens are built in the Phase 0 frontend prototype and become the real app in Phase 4.

Screen	Route	Role	FRs
Landing page	/	Public	—
Login / OTP	/login	All	FR-01
Role select + profile setup	/onboarding	Farmer, Company	FR-02, FR-03
Farmer dashboard	/farmer	Farmer	FR-08, FR-13
Crop advice	/farmer/advice	Farmer	FR-05, FR-06
My listings + create listing	/farmer/listings	Farmer	FR-07, FR-08
Listing detail with matches	/farmer/listings/[id]	Farmer	FR-08, FR-10
Offers / deals inbox	/farmer/deals	Farmer	FR-12
Market prices	/market	Farmer, Company	FR-15
Company dashboard	/company	Company	FR-10
Post requirement	/company/requirements/new	Company	FR-09
Browse produce	/company/browse	Company	FR-11
Deal detail (shared)	/deals/[id]	Farmer, Company	FR-12, FR-17
Admin overview	/admin	Admin	FR-18
Admin verifications	/admin/verifications	Admin	FR-04
Admin data & model health	/admin/system	Admin	FR-18

9. Data rules and validation

Field	Rule
Phone	Indian mobile, 10 digits, starts 6–9; stored E.164 (+91…).
OTP	6 digits; 5-minute expiry; single use.
GSTIN	15 characters, regex ^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z][1-9A-Z]Z[0-9A-Z]$; checksum verified.
Quantity	Decimal > 0; max 10,000 t per listing; units: quintal (q) or tonne (t); stored internally in kg.
Price	₹ per quintal; integer; min ≤ max for ranges.
Dates	Ready date and required-by date ≥ today; displayed DD MMM YYYY (Indian format).
Location	Lat/long required (map pin or GPS); district selected from master list.
Photos	JPG/PNG/WebP, max 3 per listing, 5 MB each, EXIF location stripped before storage.
Free text	Max 500 characters; HTML stripped.

10. AI / ML requirements (summary)

Full technical design is in the TRD. Product-level requirements:

- Forecast horizon 2–12 weeks; outputs demand level (Low/Medium/High), trend (Falling/Stable/Rising), price direction (Down/Stable/Up) and confidence (0–100%).
- Every prediction carries a model version and is presented as advisory, never as a guarantee.
- Cold-start fallback to state- or crop-level averages, clearly labelled "low confidence".
- Explanations generated from top feature contributions (SHAP) mapped to templated sentences, translated to Hindi and Marathi.
- Retraining at least monthly; drift monitoring; metrics by region and season in a model card.

11. Non-functional requirements

Area	Requirement
Usability	Mobile-first, 48 px touch targets, icons + short text, works on low-end Android (2 GB RAM).
Performance	Pages < 3 s on 4G; prediction API < 2 s p95; 1,000 concurrent users.
Availability	99% uptime; daily DB backups with tested restore.
Security	See Security & Privacy document: HTTPS, RBAC, OWASP Top 10, OTP rate limits, audit logs.
Privacy	Minimum data; consent notice; DPDP Act 2023 / DPDP Rules 2025 alignment.
Localisation	Externalised strings; Devanagari fonts; Indian date/number formats; quintal/tonne.
Maintainability	Modular code, OpenAPI docs, 70%+ unit test coverage on core logic, linting, env-based config.
Trust	Advisory disclaimer; Terms and Privacy pages; platform positioned as connector, not trading party (pending legal review).

12. Release plan

Phase	Duration	Deliverables
0. Discovery + clickable prototype	1–2 weeks	Signed PRD, TRD, clickable frontend with mock data (college demo), data-source feasibility spike.
1. Foundation	2 weeks	Repo, DB schema, OTP auth, roles, profiles, admin verification.
2. Data and AI	3 weeks	Pipelines, datasets, baseline + improved models, validation report, prediction APIs.
3. Marketplace	2–3 weeks	Listings, requirements, matching, deals, notifications.
4. Frontend wiring + i18n	2 weeks (parallel)	Prototype connected to real APIs; Hindi + Marathi.
5. Testing and pilot	2 weeks	QA, security tests, UAT with farmers and companies, performance tuning.
6. Deploy and handover	1 week	Production, docs, admin training, credentials handover.

Indicative total: 11–13 weeks for MVP including the prototype week. Weekly demo every Friday.

13. Acceptance criteria

- All Must-have FRs implemented and demonstrated.
- Staging demo script passes: farmer registers → gets crop advice → posts listing → verified company posts requirement → match suggested → offer sent → farmer accepts → remaining quantity stays listed.
- Model validation report with metrics by region and season vs naive baseline.
- No critical/high open bugs; medium bugs documented with fix plan.
- Tested on 3 Android devices (one low-end), Chrome and Safari.
- Source code in client-owned repository with README, setup guide, API docs, env config.
- Production deployment with backups and monitoring configured.

14. Assumptions, risks and open items

14.1 Key risks

Risk	Mitigation
Government data incomplete, delayed or unreachable	Phase 0 feasibility spike; use CEDA (Ashoka) historical API as backup; cache; admin CSV upload.
Low accuracy for some crops/regions	Show confidence; limit pilot to data-rich crops; always advisory.
Cold start (few users)	Seed with FPOs and local processors; admin can post requirements on behalf of companies.
Low digital literacy	Simple UI, local language, voice later, assisted onboarding by field agents.
Fake listings / disputes	Company verification, flagging, audit logs, ratings.
Scope creep	Written change requests; anything outside Section 6.1 estimated separately.

14.2 Open items for client

#	Question	Owner	Status
1	Budget, timeline, payment milestones	Onkar / Client	Open
2	Pilot district and first 3–5 crops (suggest: maize, soybean, onion, tur, tomato)	Client	Open
3	Languages beyond EN/HI/MR	Client	Open
4	Business model (free / commission / company subscription)	Client	Open
5	Legal position (connector vs trading party), Terms, Privacy Policy	Client	Open
6	Branding, domain, use of the name "AgriDirect AI"	Client	Open
7	Cloud accounts and API key ownership (SMS, maps, WhatsApp)	Client / Dev	Open
8	Existing FPO / company data for pilot	Client	Open
9	Data licensing — Agmarknet/data.gov.in (OGL India), Open-Meteo (commercial plan needed)	Developer	In research

15. Glossary

Term	Meaning
Mandi / APMC	Regulated wholesale agricultural market (Agricultural Produce Market Committee).
Modal price	Most common transaction price for a commodity in a mandi on a given day.
Arrivals	Quantity of a commodity arriving at a mandi on a given day.
FPO	Farmer Producer Organisation — collective of farmers selling together.
Quintal	100 kg. 1 tonne = 10 quintals.
Kharif / Rabi / Zaid	Monsoon (Jun–Oct), winter (Oct–Mar) and summer cropping seasons.
PWA	Progressive Web App — website installable on a phone home screen.
MAPE / RMSE / F1	Forecast error and classification quality metrics.